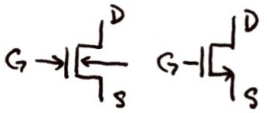
EECS311 Mid-2

MOSFET

NMOS



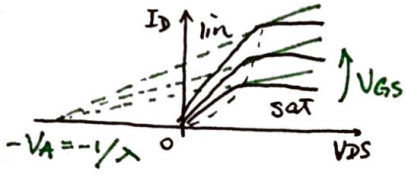
① Cutoff $V_{GS} < V_{tn}$. $I_D = 0$

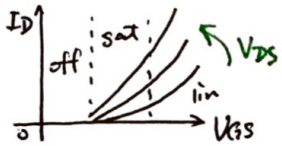
② linear $V_{GS} > V_{tn}$, $V_{DS} \le V_{GS} - V_{tn} \equiv V_{ov}$

$I_D = K_n(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$ * $K_n = K'W/L$, $K' = \mu_n C_{ox}$

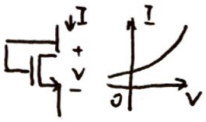
③ saturation $V_{GS} > V_{tn}$, $V_{DS} > V_{ov}$

$I_D = \frac{1}{2}K_n V_{ov}^2(1+\lambda V_{DS})$ * $\lambda = \Delta L/L$ (C.L.M)



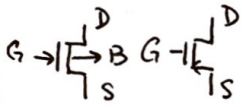


diode-connected transistor



sat $\Rightarrow I_D = \frac{1}{2}KV_{ov}^2 = \frac{1}{2}K(V-V_{tn})^2$

PMOS



① Cutoff $V_{GS} > V_{tp}$ $I_D = 0$

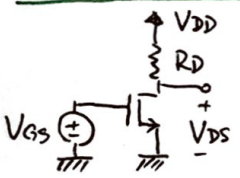
② linear $V_{GS} < V_{tp}$, $V_{DS} \ge V_{GS} - V_{tp} \equiv V_{ov}$

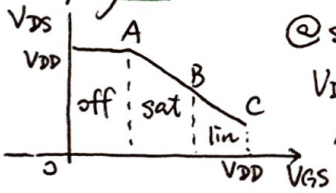
$I_D = -k_p(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

③ saturation $V_{GS} < V_{tp}$, $V_{DS} < V_{ov}$

$I_D = -\frac{1}{2}k_p V_{ov}^2(1+\lambda V_{DS})$

Common Source Amplifier



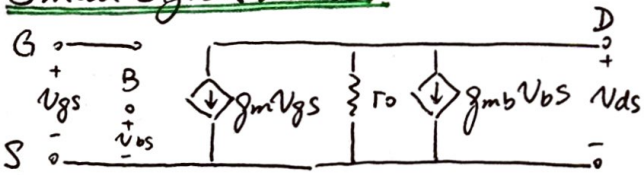


@sat

$V_{DS} = V_{DD} - \frac{1}{2}R_D K_n V_{ov}^2$

$A_v = -g_m R_D$

Small Signal Model

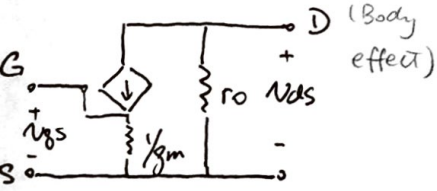


Hybrid-π model

$g_{m,sat} = \frac{\partial i_{D,sat}}{\partial v_{GS}}\Big|_{v_{GS}=V_{GS}} = K_n(V_{GS} - V_{tn}) = \sqrt{2K_n I_{D,sat}} = -\frac{2R_D I_D}{V_{ov}}$

$r_o = \left(\frac{\partial i_D}{\partial v_{DS}}\right)^{-1} = V_A/I_D$ (CLM)

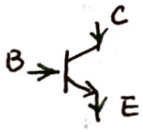
$g_{mb} = \left(\frac{\partial i_D}{\partial v_{BS}}\right) = \frac{\gamma}{2\sqrt{2\phi_F + V_{SB}}}$



T-model

BJT

NPN



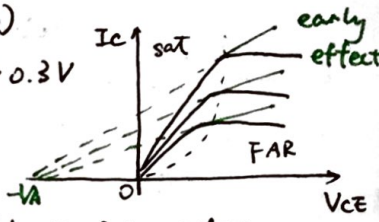
① Cutoff EB.CB reversed

② FAR EB forward, CB reversed

$\alpha = i_C/i_E$, $\beta = i_C/i_B$, $i_C = I_S e^{V_{BE}/V_T}$, $V_T = 25.8mV$

$\alpha = \beta/(\beta+1) \Leftrightarrow \beta = \alpha/(1-\alpha)$

$V_{BE} = 0.7V$, $V_{BC} < 0.4V$, $V_{CE} > 0.3V$



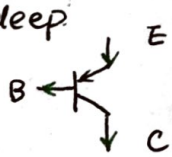
③ sat EB.CB forward

$\beta_{forced} \le \beta$. $i_C \downarrow$. $i_b \uparrow$

$V_{BE} = 0.7V$ | $V_{BC} = 0.4V$, $V_{CE} = 0.3V$ edge

| $V_{BC} = 0.5V$, $V_{CE} = 0.2V$ deep

PNP

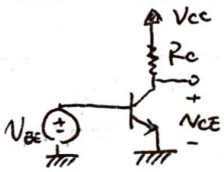


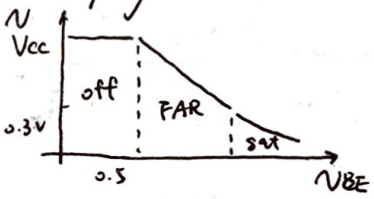
Simply swap subscripts of NPN.

Early Effect: $\beta_F = \beta_{F0}(1 + V_{CE}/V_A)$

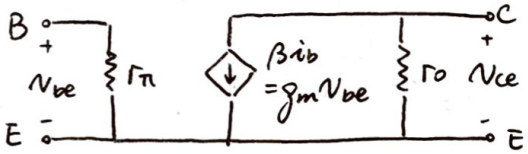
$i_C = I_S e^{V_{BE}/V_T}(1 + V_{CE}/V_A)$

Common Emitter Amplifier





Small Signal Model



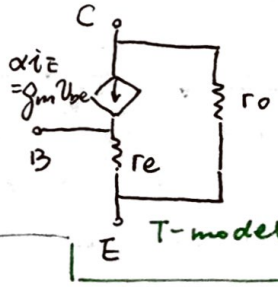
Hybrid-π model

$g_m = I_C/V_T$

$r_o = (V_A + V_{CE})/I_C = V_A/I_C$

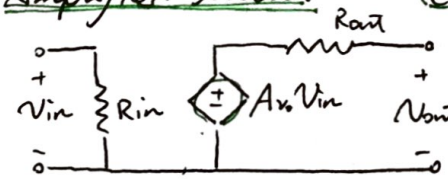
$r_\pi = \beta/g_m = V_T/I_B = \beta V_T/I_C$

$r_e = r_\pi/(\beta+1) = V_T/I_E = \alpha/g_m$



T-model

Amplifier Model



For common source amp

$R_{in} = \infty$, $R_{out} = r_o // R_D$

$A_{vo} = -g_m(R_D // r_o)$

For common emitter amp

$R_{in} = r_\pi$, $R_{out} = r_o // R_C$

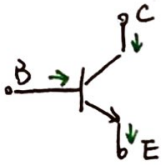
$A_{vo} = -g_m(R_C // r_o)$

Overall voltage gain

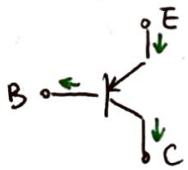
$G_v = R_{in}/(R_{in} + R_{sig}) \cdot A_{vo} \cdot R_L/(R_L + R_{out})$

Large Signal Mode of BJT

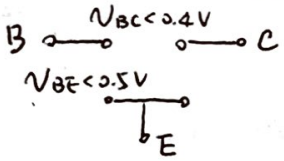
NPN

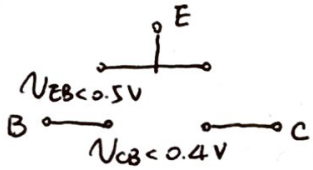


PNP

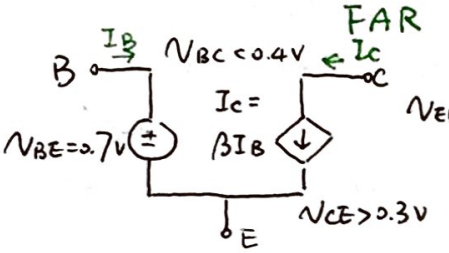


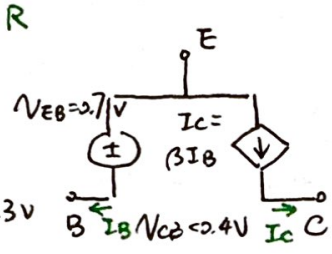
cut off



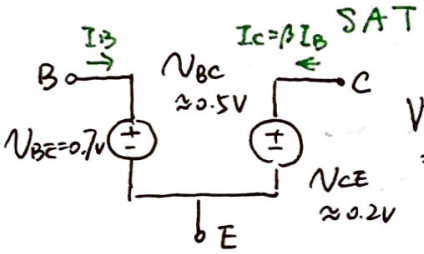


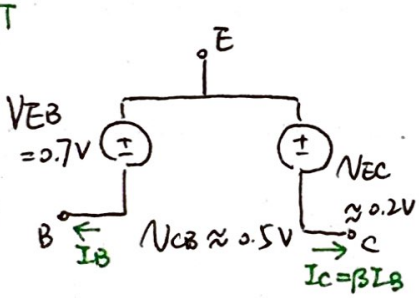
FAR



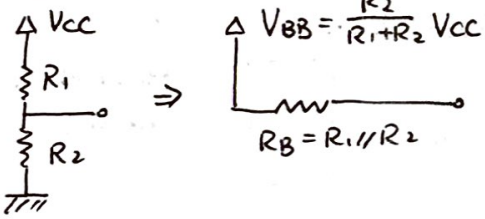


SAT

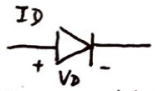


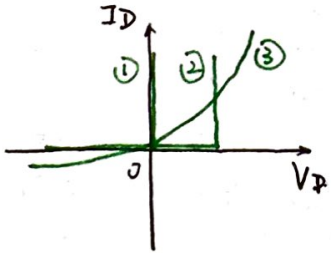


Symplified model



Diode





① ideal $\begin{cases} I_D>0,\ V_D\geq 0 \\ I_D=0,\ V_D<0 \end{cases}$

② constant voltage $\begin{cases} I_D>0,\ V_D\geq V_{ON} \\ I_D=0,\ V_D<V_{ON} \end{cases}$

③ exponential $I_D=I_S(e^{qV_D/nkT}-1)$

$=I_S e^{V/V_T}$ (if $i\gg I_S$)

$V_T=25.8mV$ (@ room tempreture)